

Curriculum-Based AI Tutor

Final Project Report — NCERT Class 8 Science

1. Project Overview

The Curriculum-Based AI Tutor is a retrieval-augmented question-answering system designed to answer student questions using NCERT Class 8 Science textbook content. The objective is to provide concise, grade-appropriate answers while restricting the knowledge source to the prepared textbook corpus. The system combines semantic search, FAISS retrieval, a language model, and a Streamlit chat interface.

The project follows a Retrieval-Augmented Generation (RAG) workflow. Relevant textbook chunks are retrieved first and supplied to the language model as context. This makes the answer generation more traceable and supports the display of textbook source snippets to the student.

2. Data Preparation and Corpus

The downloaded NCERT Class 8 Science PDF was processed in the Jupyter Notebook. The workflow included PDF text extraction, cleaning, chapter/page organization, and text chunk creation. The cleaned data was saved as `class8_science.jsonl` for indexing and later application use.

Final corpus validation produced **307 text chunks** and **18 unique chapter entries** in the prepared corpus. Chunk sizes ranged from 268 to 2,267 characters, with an average of approximately 1,393 characters.

Scope note: The project brief specifies 13 chapters, while the actual final validation shown during implementation contained 18 chapter entries. The implementation result is reported as observed; the final corpus should be checked against the exact textbook edition before submission.

3. Embeddings and Vector Index

Each corpus chunk was converted into a semantic vector using `sentence-transformers/all-MiniLM-L6-v2`. The embeddings have 384 dimensions. FAISS was used for nearest-neighbour search, with the indexed vectors associated with the prepared textbook documents.

Component	Implementation
Corpus	307 textbook chunks
Embedding model	all-MiniLM-L6-v2
Embedding dimension	384
Vector index	FAISS
Indexed vectors	307
Stored documents	307

4. RAG Pipeline and AI Tutor

For each student question, the application generates a query embedding and searches the FAISS index for the top three relevant textbook chunks. A similarity threshold helps reject queries that are not sufficiently related to the prepared textbook content. Retrieved context is then inserted into a textbook-only prompt.

The answer-generation stage uses `google/flan-t5-small`. The prompt instructs the model to use only the supplied textbook context, avoid invented facts, use simple Class 8 language, and state when sufficient textbook information cannot be found. The Streamlit interface maintains conversation history through session state and displays chapter, page, similarity score, and source snippets.

5. Evaluation

The implemented evaluation used 10 test questions. Generated answers were compared with reference answers using BLEU and ROUGE-L. A brief human review recorded correctness and relevance. The observed results were:

Evaluation measure	Result
Number of test questions	10
Average BLEU	0.0116
Average ROUGE-L	0.0962
Average correctness	4.0 / 5
Average relevance	5.0 / 5
Average semantic similarity	0.5529

6. Results and Error Analysis

The human-review results indicate that the tested answers were generally relevant and reasonably correct when suitable textbook context was retrieved. The average relevance score was 5.0/5 and the average correctness score was 4.0/5.

The relatively low BLEU (0.0116) and ROUGE-L (0.0962) scores indicate limited lexical overlap between generated and reference answers. A semantically correct answer can use different wording, so these metrics should be interpreted together with human review and retrieval similarity. The results also suggest that the evaluation set, prompting, retrieval strategy, and generation model can be improved.

An error-analysis stage was completed and interactions were logged for later inspection. The logs can help identify weak retrieval, low-overlap answers, and questions requiring better prompt design or corpus coverage.

7. Streamlit Deployment

A single-page Streamlit application was developed as the demonstration interface. It provides a dark-themed chat dashboard, session-based conversation history, semantic retrieval, generated answers, textbook source snippets, similarity information, and an out-of-syllabus fallback. Interactions are recorded for later error analysis.

8. Conclusion

The project demonstrates a complete curriculum-focused RAG pipeline: textbook extraction and cleaning, semantic embeddings, FAISS indexing, contextual retrieval, constrained answer generation, evaluation, logging, and Streamlit deployment. The approach provides a practical educational assistant whose answers can be traced to retrieved textbook content.

9. Future Work

- Verify and align the final corpus with the exact 13-chapter scope specified in the project brief.
- Expand evaluation from 10 to 20 diverse questions, covering definitions, explanations, comparisons, and application-style questions.
- Improve retrieval through better chunking, metadata filtering, and retrieval-quality evaluation.
- Experiment with a stronger instruction-tuned model while retaining textbook-only constraints.
- Apply the optional LoRA fine-tuning stage using curated question-answer pairs if additional alignment is needed.
- Strengthen out-of-syllabus detection and safeguards against unsupported answers.

Final deliverables: ai_tutor_class8.ipynb, class8_science.jsonl, evaluation.csv, app.py, interaction/error-analysis logs, and this final report.